INCH-POUND
A-A-55630
5 December 1997

COMMERCIAL ITEM DESCRIPTION

FILLERS, CONTINUOUS, FOR ROTARY SWEEPER BRUSH: FOR SWEEPER, FRONT MOUNTED OR TOWED TYPE

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID covers the general requirements for continuous, flat and convoluted ring, steel and plastic fillers for high speed rotary sweepers. Fillers covered by this CID are intended for commercial/industrial applications.

2. CLASSIFICATION. The continuous fillers shall be of the following types and classes (see 7.2 and 7.4):

Type 01 - Flat ring, wire filler (Regular Duty)
Type 02 - Flat ring, wire filler (Heavy Duty, Class A only)
Type 03 - Flat ring, plastic filler
Type 11 - Convoluted ring, wire filler (Regular Duty, Class A only)
Type 12 - Convoluted ring, wire filler (Heavy Duty, Class A only)
Type 13 - Convoluted ring, plastic filler (Class A only)

Class A - 10.75 inch Inside Diameter
Class B - 19.50 inch Inside Diameter

3. SALIENT CHARACTERISTICS.

3.1 Environmental conditions. The fillers shall satisfactorily withstand storage temperatures ranging from -60°F to +160°F and operating temperatures ranging from -40°F to +125°F.

3.2 Operating conditions. The fillers shall be for use on sweepers that operate at 0-800 rpm broom core rotational speed.

3.3 Performance. When operated on concrete runways at a sweeping speed of 35 mph, a rotational speed of 500 rpm and with a brush pattern of not less than 2 inches but not greater than 4 inches, a new filler shall be capable of operating satisfactorily for a minimum of 160 hours. A lack of water/snow, as well as runway surfaces that have been milled for run off grooves may reduce filler operating time to 128 hours. For wire fillers the loss of wire 2 inches or longer throughout the life of the filler shall not exceed 2 percent of the overall initial weight of the filler. The fillers shall maintain this performance at any operational temperature in the range specified in 3.1.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Industrial Supply Center, Attn.: DISC-BBEE, 700 Robbins Avenue, Philadelphia, PA 19111-5096.

AMSC N/A

FSC 3825

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3.4 Construction. Fillers shall have an inside diameter of 10.75 -0 + .125 inches for Class A and 19.50 -0 + .125 inches for Class B. Overall outside diameter shall be 36 ± .50 inches for Class A and 46 ± .50 inches for Class B. Each filler shall consist of a steel ring filled with zinc coated wire or polypropylene (poly) bristles and shall be of the type and class as specified in section 2. Each filler shall be marked with a black dot or other suitable marking on the ring to indicate the point of maximum static unbalance.

3.4.1 Ring. The ring shall be made of coil steel, minimum thickness .040 inch, conforming to SAE J403 (UNS G10180) and shall be solid and continuous with all joints welded to insure structural integrity. The ring shall be either the flat or convoluted type as specified in section 2. Each ring shall have steel drive pins to engage the sweeper core. Class A pins shall have a minimum diameter of .313 inch. Class B pins shall have a minimum diameter of .250 inch. The length of the pins and number of pins per wafer is specified below under the individual wafer types. Each ring shall be pressed closed every 1 ± .125 inch to ensure maximum strength under high speed conditions.

3.4.2 Bristles.

3.4.2.1 Wire Bristles. The wire bristles shall be made of steel conforming to SAE J403 (UNS G10700) with a regular zinc coating. The bristles shall be .018 inch diameter minimum. The bristles shall be crimped with not less than 3 crimps per inch at a crimp amplitude of .063 inch minimum. The regular zinc coating shall be applied at not less than .01 ounce per square foot and the tensile strength of the wire shall be in the range of 310,000 - 355,000 pounds per square inch (psi) after crimping. The bristles shall be held secure in the ring by not less than 3 wraps of .090 inch diameter minimum or 5 wraps of .076 inch diameter minimum zinc coated UNS G10080 to UNS G10700 steel retaining wire.

3.4.2.2 Plastic Bristles. Plastic bristles shall be formed of virgin polypropylene strands and have an oval cross section not less than .070 x .105 inch. The strands shall be crimped and the tensile strength shall be 35,000 psi minimum after crimping. The ring shall have a rubber liner to absorb friction and protect the bristles from abrasion at the ring edges. The bristles shall be held secure in the ring by not less than 3 wraps of .090 inch diameter minimum or 5 wraps of .076 inch diameter minimum zinc coated UNS G10080 to UNS G10700 steel retaining wire.

3.4.3 Filler types.

3.4.3.1 Type 01 - Flat ring, wire filler (Regular Duty). The filler shall be a flat ring filled with wire bristles as specified in 3.4.2.1. The Class A filler shall have one drive pin protruding .500 to .563 inch from the ring. The Class B filler shall have not have less than 4 drive pins. Each pin shall protrude at least .250 inch from the ring. The pins shall be 15.500 ± .063 inches apart center to center. The weight of the complete filler shall be 5.50 ± .25 lbs. for Class A and 8.50 ± .50 lbs. for Class B.

3.4.3.2 Type 02 - Flat ring, wire filler (Heavy Duty, Class A only). This filler shall be identical to the Class A filler described in 3.4.3.1 except that the bristles shall be held in the ring by not less than 4 wraps of .090 inch diameter minimum or 8 wraps of .076 inch diameter minimum retaining wire and the weight of the complete filler shall be 8.00 ± .40 lbs.

3.4.3.3 Type 03 - Flat ring, plastic filler. The filler shall be a flat ring filled with plastic bristles as specified in 3.4.2.2. The Class A filler shall have one drive pin protruding .500 to .563 inch from the ring. The Class B filler shall have not less than 4 drive pins. Each pin shall protrude at least .250 inch from the ring. The pins shall be 15.500 ± .063 inches apart center to center. The weight of the complete filler shall be 5.00 ± .25 lbs. for Class A and 8.00 ± .40 lbs. for Class B.

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3.4.3.4 Type 11 - Convoluted ring, wire filler (Regular Duty, Class A only). The filler shall be a convoluted ring filled with wire bristles as specified in 3.4.2.1. The filler shall have at least 2 drive pins protruding .500 to .563 inch from the ring and spaced 1.438 to 1.563 inches apart center to center. Weight of the filler shall be 5.50 ± .25 lbs.

3.4.3.5 Type 12 - Convoluted ring, wire filler (Heavy Duty, Class A only). This filler shall be identical to the filler described in 3.4.3.4 except that the bristles shall be held in the ring by not less than 4 wraps of .090 inch diameter minimum or 8 wraps of .076 inch diameter minimum retaining wire and the weight of the complete filler shall be 8.00 ± .40 lbs.

3.4.3.6 Type 13 - Convoluted ring, plastic filler (Class A only). The filler shall be a convoluted ring filled with plastic bristles as specified in 3.4.2.2. The filler shall have at least 2 drive pins protruding .500 to .563 inch from the ring and spaced 1.438 to 1.563 inches apart center to center. The weight of the complete filler shall be 5.00 ± .25 lbs.

4. REGULATORY REQUIREMENTS.

4.1 Recovered material. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Bid sample criteria. When specified in the contract or purchase order, a bid sample shall be inspected to determine compliance with the requirements specified in section 3.

5.3 Market acceptance. The following market acceptance criteria are necessary to document the quality of the product to be provided under this CID.

5.3.1 The company producing the item must have been producing a product meeting the requirements of this CID for at least 18 months.

5.3.2 The company must have sold 2,500 units meeting the requirements of this CID in the commercial marketplace over the past 12 months.

6. PACKAGING. Preservation, packing and marking shall be as specified in the contract or purchase order.

7. NOTES.

7.1 Addresses for obtaining copies of referenced documents.

7.1.1 Non-Government publications. SAE J403 "Chemical Compositions of SAE Carbon Steels" is available from the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001.

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7.2 Ordering data. The contract or order should specify the following:

a. CID document number and revision.

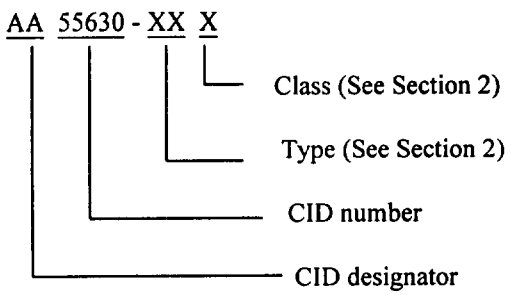
b. CID PIN.

c. Quality assurance provisions.

d. Packaging requirements.

7.3 Government Procurements. Offerors of these fillers shall identify the actual manufacturer's part number reference. This reference will be used for identification of the product offered. It is the responsibility of the offeror to ensure the product offered conforms to the requirements of this CID.

7.4 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor:



7.5 These fillers are to be used with A-A-55632 Type 04 and 05 spacers.

MILITARY INTERESTS:

Custodians:
Air Force - 99
Army - AT

Review Activity:
Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITY

GSA - FSS

Preparing Activity:
DLA - IS

Project 3825-0241

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